Hacon-Popa-Schnell's Reading Notes Fall 2025

Note III.3 — 2025 10 24 (draft round 1)

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The aim of this note is to prove the Iitaka conjecture when the base has maximal Albanese dimension. The main reference for these notes is [HPS18].

Theorem 0.1 ([HPS18, Theorem 1.1])**.** Let $f : X \to Y$ be an algebraic fiber space with general fiber F. Assume that Y has maximal Albanese dimension, then $\kappa(X) \geq \kappa(F) + \kappa(Y)$.

There are three essential ingredients that appear in the proof:

(a) The Green–Lazarsfeld–Simpson structure theorem on cohomological support loci and the generic vanishing theorem;
(b) Hacon–Popa–Schnell's construction of singular Hermitian metrics on $f_*\omega_X^{\otimes n}$;
(c) Kawamata's proof of the Iitaka conjecture when the base is of general type, together with his structure theorem for finite covers of Abelian varieties.

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1 Green-Lazarsfeld's Generic Vanishing theorem

In this section, we will introduce Green-Lazarsfeld's generic vanishing theorem. To keep the note to a reasonable length, we only give a brief overview of the results and all the proofs in the Section can be found in my note A brief introduction to generic vanishing theorem.

1.1 Theorem of Hacon-Pareschi-Popa and GV sheaves

Definition 1.1 (GV (generic vanishing) sheaves)**.** Given a coherent $\mathscr{O}_T$-module $\mathscr{F}$ on a compact complex torus T, define

$$S^i(T, \mathscr{F}) = \left\{L \in \operatorname{Pic}^0(T) \mid H^i(T, \mathscr{F} \otimes L) \neq 0\right\}$$

We say that $\mathscr{F}$ is a GV-sheaf if $\operatorname{codim} S^i(T, \mathscr{F}) \geq i$ for every $i \geq 0$; we say that $\mathscr{F}$ is M-regular if $\operatorname{codim} S^i(T, \mathscr{F}) \geq i+1$ for every $i \geq 1$.

Definition 1.2 (Cohomological support loci)**.** Recall that for any coherent sheaf $\mathscr{F}$ on an abelian variety A, we consider for all $k \geq 0$ the cohomological support loci

$$V^k(\mathscr{F}) = \left\{P \in \operatorname{Pic}^0(A) \mid H^k(A, \mathscr{F} \otimes P) \neq 0\right\}$$

They are closed subsets of $\operatorname{Pic}^0(A)$, by the semi-continuity theorem for cohomology.

Definition 1.3 (Unipotent Vector Bundle)**.** A vector bundle on A is called unipotent if it has a filtration

$$0 = U_0 \subset U_1 \subset \cdots \subset U_n = U$$

such that $U_i/U_{i-1} \simeq \mathscr{O}_A$ for all $i = 1, \ldots, n$. Note in particular that $\det U \simeq \mathscr{O}_A$.

That is, we have a successive extension

$$0 \to U_{i-1} \to U_i \to E_i \to 0$$

with

$$U_i/U_{i-1} \cong \mathcal{O}_A = E_i$$

Before going into the proof of the theorem of Hacon–Pareschi–Popa, let us briefly introduce the Fourier-Mukai Transform.

Definition 1.4 (Fourier-Mukai Transform)**.** Let A be an Abelian variety. We will identify $\operatorname{Pic}^0(A)$ with the dual abelian variety $\widehat{A}$, and denote by P the normalized Poincaré bundle on $A \times \widehat{A}$. It induces the integral transforms

$$RS : D^b_{coh}\left(\mathscr{O}_A\right) \longrightarrow D^b_{coh}\left(\mathscr{O}_{\widehat{A}}\right), \quad RS\mathscr{F} = \mathbf{R}p_{2_*}\left(p_1^*\mathscr{F} \otimes P\right).$$

and

$$R\hat{S} : D^b_{coh}\left(\mathscr{O}_{\widehat{A}}\right) \longrightarrow D^b_{coh}\left(\mathscr{O}_A\right), \quad R\hat{S}\mathscr{G} = \mathbf{R}p_{1_*}\left(p_2^*\mathscr{G} \otimes P\right),$$

where $p_1 : A \times \hat{A} \to A$ and $p_2 : A \times \hat{A} \to \hat{A}$. (Note that in some of the references, the Fourier-Mukai transform $R\Psi_P = R\hat{S}$ and $R\Phi_P = RS$).

The following result of Mukai is fundamental.

Theorem 1.5. The Fourier-Mukai functors $(\mathbf{R}S, \mathbf{R}\hat{S})$ are equivalences of derived categories, usually called the Fourier-Mukai transforms; moreover,

$$\mathbf{R}\hat{S} \circ \mathbf{R}S = (-1_A)^* [-g] \quad \text{and} \quad \mathbf{R}S \circ \mathbf{R}\hat{S} = \left(-1_{\hat{A}}\right)^* [-g]$$

where $[-g]$ denotes shifting g places to the right.

We will give the proof of Mukai's theorem in the Series of notes A brief introduction to Fourier-Mukai Transform.

One can prove the following commutation relation of Fourier-Mukai Transform and Grothendieck Dual.

Proposition 1.6 (Commutation relation of Grothendieck dual and Fourier-Mukai Transform)**.**

$$D_A \circ RS = \left((-1_A)^* \circ RS \circ D_{\hat{A}}\right) [g].$$

Using the Fourier–Mukai transform, the following non-vanishing theorem comes for free. (This shows that to get non-vanishing of $h^0(\mathscr{F})$ one needs a lot of vanishing on h^i for $i > 0$).

Proposition 1.7. If $\mathcal{F} \in \mathrm{Coh}(A)$ is a non-zero coherent sheaf such that $h^i(\mathcal{F} \otimes P) = 0$ for all $i > 0$ and $P \in \hat{A}$, then $h^0(\mathcal{F}) \neq 0$.

Hacon-Pareschi-Popa proved the following equivalence characterization of GV sheaves.

Theorem 1.8 (Theorem of Hacon-Pareschi-Popa, [HPS18, Lemma 7.3 + Theorem 7.7])**.** Let A be an Abelian variety of dimension g over $\mathbb{C}$, $\mathscr{F}$ be a coherent sheaf on A. Then the following are equivalent.

(1) (Vanishing condition) $h^i(A, \mathscr{F} \otimes \hat{L}^\vee) = 0$ for any $i > 0$. Here L is a sufficiently ample line bundle on $\hat{A}$, and $\hat{L}$ is the vector bundle on A that you get by Fourier-Mukai transform $\hat{L} = R^0 S(L)$;

(2) (The Fourier-Mukai condition)

$$R\hat{S}(D_A(\mathscr{F})) = R^g \hat{S}(D_A(\mathscr{F})),$$

where D_A is the Grothendieck derived dual and $R\hat{S}$ denotes the Fourier-Mukai transform.

(3) (Codimension condition on cohomological support loci)

$$\operatorname{codim} V^i(\mathscr{F}) \geq i, \quad \forall i > 0$$

For GV sheaves, the cohomological support loci satisfy a certain order relation.

Theorem 1.9. Let $\mathscr{F}$ be a coherent sheaf on A. Then if $\mathscr{F}$ is a GV-sheaf, then

$$V^g(\mathscr{F}) \subseteq \cdots \subseteq V^1(\mathscr{F}) \subseteq V^0(\mathscr{F}).$$

As an immediate consequence, we can get a non-vanishing criterion for GV sheaves.

Corollary 1.10 ([HPS18, Lemma 7.4])**.** Let A be an Abelian variety, $\mathscr{F}$ be a GV sheaf on A. Then

$$\mathscr{F} = 0 \iff V^0(\mathscr{F}) = \emptyset.$$

Using the theorem of Hacon-Pareschi-Popa we can prove the direct image of pluricanonical sheaves are GV sheaves.

Proposition 1.11. Let $f : X \to A$ be a morphism from a smooth projective variety to an abelian variety. For every $m \in \mathbb{N}$ and $i \geq 1$, the sheaf $\mathscr{F}_m = R^i f_* \omega_X^{\otimes m}$ is a GV-sheaf on A.

1.2 Green-Lazarsfeld generic vanishing theorem

Let us first recall the Green–Lazarsfeld–Simpson generic vanishing theorem. We will begin by introducing Green and Lazarsfeld's original proof, and then present Hacon's proof of the Green–Lazarsfeld theorem using the Fourier–Mukai transform.

Theorem 1.12. If $f : X \to A$ is a morphism from a smooth projective variety to an abelian variety, then for any $j, k \geq 0$ we have
(1) $\operatorname{codim}_{\operatorname{Pic}^0(A)} V^k\left(R^j f_* \omega_X\right) \geq k$;
(2) Every irreducible component of $V^k\left(R^j f_* \omega_X\right)$ is a translate of an abelian subvariety of A by a point of finite order.

For our interest, we need the generic vanishing for direct image of pluricanonical sheaves.

Theorem 1.13 ([HPS18, Theorem 4.1])**.** Let X be a smooth projective variety. For each $m \in \mathbb{N}$, the locus

$$\left\{P \in \operatorname{Pic}^0(X) \mid H^0\left(X, \omega_X^{\otimes m} \otimes P\right) \neq 0\right\} \subseteq \operatorname{Pic}^0(X)$$

is a finite union of abelian subvarieties translated by points of finite order.

This theorem implies that $V^0\left(A, \mathscr{F}_m\right)$ (for $\mathscr{F}_m = f_* \omega_X^{\otimes m}$) is also a finite union of abelian subvarieties translated by points of finite order, where

$$\begin{aligned} V^0\left(A, \mathscr{F}_m\right) &= \left\{P \in \operatorname{Pic}^0(A) \mid H^0\left(A, \mathscr{F}_m \otimes P\right) \neq 0\right\} \\ &= \left\{P \in \operatorname{Pic}^0(A) \mid H^0\left(X, \omega_X^{\otimes m} \otimes f^* P\right) \neq 0\right\} \subseteq \operatorname{Pic}^0(A). \end{aligned}$$

Using the result we just developed above, we can prove the following criterion for direct image of pluricanonical sheaves to be an indecomposable unipotent vector bundle.

Theorem 1.14 ([HPS18, Corollary 4.3])**.** Let X be a smooth projective variety with $\kappa(X) = 0$, and let $f : X \to A$ be an algebraic fiber space over an abelian variety.

If $H^0\left(X, \omega_X^{\otimes m}\right) \neq 0$ for some $m \in \mathbb{N}$, then the coherent sheaf $\mathscr{F}_m$ is an indecomposable unipotent vector bundle.

2 Hacon-Popa-Schnell's construction of semi-positive singular Hermitian metric on $f_* \omega_X^{\otimes n}$

The technical core of the result (Theorem 0.1) lies in the construction of the semi-positive metric on the direct image sheaf $f_* \omega_X^{\otimes n}$ that we will construct in this section.

We will divide the construction into 3 steps. First we try to find some semi-positive singular Hermitian metric on $\mathscr{F} = f_*(\omega_{X/Y} \otimes L \otimes \mathcal{I}(h))$ as long as (L, h) is semi-positive using the optimal Ohsawa-Takegoshi extension theorem. In the second step, we try to construct the m-th Narasimhan-Simha metric on $\omega_{X/Y}$ and hence a semi-positive line bundle $(\omega_{X/Y}^{\otimes(m-1)}, h)$. And finally, using the elimination of multiplier ideal lemma, we can transport the semi-positivity from $\mathscr{F} = f_*(\omega_{X/Y} \otimes L \otimes \mathcal{I}(h))$ to $\mathscr{F}_m = f_*(\omega_{X/Y}^{\otimes m})$. It is worth mentioning that up to now there is no algebraic proof of such a semi-positivity theorem.

2.1 Construction of semi-positive metric on $f_*(\omega_{X/Y} \otimes L \otimes \mathcal{I}(h))$

We first prove there exists a semi-positive singular Hermitian metric on $\mathscr{F} = f_*\left(\omega_{X/Y} \otimes L \otimes \mathcal{I}(h)\right)$ as long as (L, h) is semi-positive.

Proposition 2.1 ([HPS18, Theorem 21.1])**.** Let $f : X \to Y$ be a projective surjective morphism between two connected complex manifolds. If (L, h) is a holomorphic line bundle with a singular hermitian metric of semi-positive curvature on X, then the pushforward sheaf

$$\mathscr{F} = f_*\left(\omega_{X/Y} \otimes L \otimes \mathcal{I}(h)\right)$$

(1) has a canonical singular hermitian metric H;
(2) This metric has semi-positive curvature and;
(3) satisfies the minimal extension property.

PROOF IDEA 2.2. The construction of the metric is a bit involved. Let us briefly sketch the idea of the proof first. We try to first construct some semi-positive singular Hermitian metric on the "nice" locus where the direct image is a vector bundle. And then we try to apply the Riemann extension theorem to extend such semi-positive metric to semi-positive metric for torsion free sheaf $f_*(\omega_{X/Y} \otimes L \otimes \mathcal{I}(h))$. Now let's explain the idea of the proof step by step.

Step 1. (Find the nice locus on which $\mathscr{F}$ is a locally free sheaf E). We first find a nice Zariski open subset $U = Y \setminus Z$, such that: (a) f is a submersion on U (so that Ehresmann's theorem can apply), (b) the direct image $\mathscr{F} = f_*(\omega_{X/Y} \otimes L \otimes \mathcal{I}(h))$ and $f_*(\omega_{X/Y} \otimes L)$ are locally free on U; (c) the base change property holds for $f_*(\omega_{X/Y} \otimes L)$.

Step 2. (Construction of the singular metric H on E). Let $s \in E(y) \subset H^0(X_y, \omega_{X_y} \otimes L_y)$ for some $y \in U$, we define the singular Hermitian inner product on E_y as

$$\|s\| = \int_{X_y} |s|_h^2.$$

We will check that this is well-defined and finite almost everywhere, hence defines a singular Hermitian metric on E.

Step 3. (Prove the semi-positivity of the metric H using optimal Ohsawa-Takegoshi extension). By definition, in order to prove the semi-positivity of the metric on E we just defined, we need to show the plurisubharmonicity of the function $\psi(y) = \log |g|_{H^*}$ for local section $g \in H^0(U, E^*)$. To do this, we need to verify the upper semi-continuity and mean value inequality on the nice locus. Let us briefly sketch the idea for the proof of the mean-value inequality. To do

this we first simplify the problem onto the unit ball $i : B \hookrightarrow Y$ (which sends $0 \mapsto y$). By optimal Ohsawa-Takegoshi extension, for any $\alpha \in E_y$, there exists a section $s \in H^0(B, i_*\mathscr{F})$ such that

$$s(0) = \alpha, \frac{1}{\mu(B)} \int_B |s|_H^2 d\mu \leq 1.$$

And therefore, by the extremal characterization of dual metric H^* we have

$$|g|_{H^*} = \sup_{s \in E_y, \|s\|_{H,y} \leq 1} \|\langle g_y, s\rangle\| \geq |\langle g_y, s/\|s\|_{H,y}\rangle|.$$

And therefore taking log, we get

$$2\psi(y) \geq \log |\langle g_y, s\rangle|^2 - \log \|s\|_{H,y}^2$$

And therefore, by integration

$$\frac{1}{\pi} \int_\Delta 2\psi d\mu \geq \frac{1}{\pi} \int_\Delta \log |g(s)|^2 d\mu - \frac{1}{\pi} \int_\Delta \log |s|_H^2 d\mu.$$

For the 2nd term on the RHS, we apply Jensen's inequality and the optimal L^2 estimate. For the 1st term on the RHS, since $g(s)$ defines a holomorphic function we can apply the mean value inequality, which completes the proof of the mean value inequality for ψ.

As a side remark, the optimal L^2 estimate plays a crucial role in the argument above.

Step 4. (Extend the semi-positive metric onto the whole torsion free sheaf). In order to extend the metric onto $\mathscr{F}$, we use the Riemann extension theorem, and thus need to check the uniform boundedness of the psh function $\psi(y)$ on Y.

2.2 Construction of Narasimhan-Simha metric on $\omega_{X/Y}$

Our next goal is to construct the m-th Narasimhan-Simha metric on $\omega_{X/Y}$.

Proposition 2.3. Let $f : X \to Y$ be a surjective projective morphism with connected fibers between two complex manifolds. Suppose that $f_*\omega_{X/Y}^{\otimes m} \neq 0$ for some $m \geq 2$.

The line bundle $\omega_{X/Y}$ has a canonical singular hermitian metric with semipositive curvature, called the m-th Narasimhan-Simha metric. This metric is continuous on the preimage of the smooth locus of f.

PROOF IDEA 2.4. Let us first briefly sketch the idea of the proof; the idea is very similar to the construction of the metric on $f_*(\omega_{X/Y} \otimes L \otimes \mathcal{I}(h))$: we first construct a relative Bergman-type metric on the smooth locus, then compute the local weight function and prove it is a psh function, reducing the problem to the mean value inequality so that one can apply the optimal L^2 estimate. To be more concrete:

Step 1. Construction of absolute and relative version of NS metric. We first define a length function: given a pluricanonical section $v \in H^0(X, \omega_X^{\otimes m}) = V_m$, we define the length function

$$\ell(v) = \left(\int_X (c_n^m v \wedge \bar{v})^{1/m} \right)^{m/2},$$

where $c_n = 2^{-n}(-1)^{n^2/2}$. We then define the singular Hermitian metric on ω_X fiberwise: given an element $\xi \in \omega_X(x)$ (where $\omega_X(x)$ is the fiber of ω_X at $x \in X$), we set

$$|\xi|_{h_m,\omega_X,x} = \inf\left\{\ell(v)^{1/m} \mid v \in V_m \text{ satisfies } v(x) = \xi^{\otimes m}\right\} \in [0, +\infty].$$

This will induce a singular Hermitian metric on ω_X.

We then construct the relative version of the NS metric. First, given $v \in V_m = H^0\left(X, \omega_X^{\otimes m}\right)$ and a point $y \in Y$, we have

$$v|_{X_y} = v_y \otimes (dt_1 \wedge \cdots \wedge dt_r)^{\otimes m}$$

for a unique m-canonical form $v_y \in H^0\left(X_y, \omega_{X_y}^{\otimes m}\right) = V_{m,y}$. We then define the length of v_y as

$$\ell_y(v_y) = \left(\int_{X_y} (c_n^m v_y \wedge \bar{v_y})^{1/m}\right)^{m/2}.$$

Given an element $\xi \in \omega_X(x)$ (with $f(x) = y$), we set the norm

$$|\xi|_{h_m,\omega_{X/Y},x} = \inf\left\{\ell_y(v_y)^{1/m} \mid v \in V_m \text{ satisfies } v(x) = \xi^{\otimes m}\right\}.$$

One can check that this defines a singular Hermitian metric on $\omega_{X/Y}$.

One of the key points in the construction is the extremal characterization of the local weight $\psi(x)$ on ω_X and local weight $\varphi_m(x)$ on $\omega_{X/Y}$.

(Extremal Characterization of ψ_m on ω_X) Let $\sigma = dz_1 \wedge dz_2 \wedge \cdots \wedge dz_n$ be the local frame on $V \subset X$. Then the local weight

$$\psi_m = -\log|\sigma|^2_{h_m,\omega_X}.$$

A section $v \in V_m$ which under the local frame can be written as $v|_U = g_v s_0^{\otimes m}$ satisfies the extremal characterization

$$\boxed{\psi_m(x) = \frac{2}{m}\sup\left\{\log|g_v(x)| \mid v \in V_m \text{ satisfies } \ell(v) \le 1\right\}.}$$

(Extremal Characterization of φ_m on $\omega_{X/Y}$) Let s_0 (a nowhere vanishing local section) be the local frame of $\omega_{X/Y}$ over $U \subset X$; then the local weight is defined to be

$$\varphi_m = -\log|s_0|^2_{h_m,\omega_{X/Y}}.$$

Let $v \in V_m = H^0(X, \omega_X^{\otimes m})$; then it can be written locally on U as $v|_U = g_v \cdot (s_0 \wedge dt_1 \wedge \cdots \wedge dt_r)^{\otimes m}$. Using the OT extension, we have the following extremal characterization of the metric weight φ_m:

$$\boxed{\varphi_m(x) = \frac{2}{m}\sup\left\{\log|g_v(x)| \mid v \in V_m \text{ satisfies } \ell(v) \le C \text{ and } \ell_y(v_y) \le 1\right\}.}$$

Step 2. Prove the continuity of the metric on the smooth locus. We will prove it locally on U. Let $x_k \to x$ be a sequence in U converging to x. If $\lim \varphi_m(x_k) \le \varphi_m(x)$ and $\varphi_m(x) \le \liminf_{k\to\infty} \varphi_m(x_k)$, then φ_m is continuous at x. Both sides follow by the extremal characterization of the metric weight φ_m and a sequential argument.

Step 3. Prove the semi-positivity of the absolute NS metric. We first find some local coordinate under which the line bundle ω_X can be trivialized (under the local frame s_0). The metric weight can then be written as

$$\varphi_m = -\log |s_0|^2_{h_m} : U \to [-\infty, +\infty).$$

Any global section $v \in V_m = H^0(X, \omega_X^{\otimes m})$, under the local frame, can be expressed as

$$v|_U = g_v s_0^{\otimes m}.$$

Then, using again the extremal characterization, we show that

$$\varphi_m(x) = \frac{2}{m} \sup \{\log |g_v(x)| \mid v \in V_m \text{ satisfies } \ell(v) \leq 1\}.$$

Since $\log |g_v(x)|$ are psh functions, and using the upper envelope of an equicontinuous family of plurisubharmonic functions, we show that φ_m is psh and hence h_m is a semi-positive metric on ω_X.

Step 4. Prove the existence of optimal extension with respect to the length ℓ. The proof is tricky, and applies the optimal L^2 extension theorem repeatedly to construct a limiting section. To be more precise, we will first find an extension v; the section will then define a semi-positive metric on $\omega_X^{\otimes m}$ and also ω_X^{m-1}. We then apply the optimal OT extension to the adjoint bundle $\omega_X \otimes \omega_X^{\otimes(m-1)}$. The key point is that by Hölder's inequality, we can get the following monotonicity inequality:

$$\boxed{\frac{\ell\left(v^{(k+1)}\right)}{\mu(B)^{m/2}} \leq \left(\frac{\ell(v^{(k)})}{\mu(B)^{m/2}}\right)^{(m-1)/m}.}$$

If all sections $v^{(k)}$ satisfy $\ell(v^{(k)}) > \mu(B)^{m/2}$, then the limiting section must satisfy the estimate $\lim_{k\to+\infty} \ell(v_k) = \mu(B)^{m/2}$, by the monotonicity inequality.

Step 5. Proof of semi-positivity of the relative NS metric on the smooth locus. We only need to prove the local weight function of the metric is plurisubharmonic. It is therefore sufficient to prove continuity of the metric and the mean value inequality for the metric weight function. We have already proved the continuity of the NS metric in Step 2. Thus we only need to prove the mean value inequality. To do this we first prove the optimal L^2 estimate for the length: for each $u \in H^0\left(X_0, \omega_{X_0}^{\otimes m}\right)$, there exists some $v \in H^0\left(X, \omega_X^{\otimes m}\right)$ with

$$\boxed{\ell(v) \leq \mu(B)^{m/2} \cdot \ell_0(u) \text{ (CORE)}}$$

We then use the extremal definition to prove

$$\varphi_m(x) \geq \frac{2}{m} \log |g_v(x)| - \log \ell_y\left(v_y\right)^{2/m},$$

and then taking integration we get

$$\frac{1}{\pi}\int_B \varphi_m(i(y))d\mu \geq \frac{1}{\pi}\int_B \frac{2}{m}\log |g_v(i(y))|\, d\mu - \frac{1}{\pi}\int_B \log \ell_y\left(v_y\right)^{2/m} d\mu,$$

and therefore applying the mean value inequality to the 1st term on the RHS and Jensen's inequality with the optimal estimate to the 2nd term on the RHS yields the mean value inequality for φ_m.

Step 6. Extension of the semi-positive relative NS metric across the singular locus. To prove that it is possible to extend across the singular locus, we apply the Riemann extension theorem, and therefore only need to prove the local uniform boundedness of the local weight φ_m. By the extremal characterization of $\varphi_m(x)$, for every $x \in X$ there exists some $v \in H^0(X, \omega_X^{\otimes m})$ with $\ell(v) \leq C$ such that

$$\varphi_m(x) = \frac{2}{m} \log |g_v(x)|$$

with $\ell(v) \sim \int |g_v|^{2/m} d\mathrm{vol}$. We then use the mean value inequality combined with Jensen's inequality to prove

$$\varphi_m(x_0) \leq \frac{1}{\mathrm{vol}(B_R)} \int_{B_R} \frac{2}{m} \log |g_v(x)| d\mathrm{vol} \lesssim_R \ell(v) \leq C.$$

Thus φ_m is locally uniformly bounded, depending only on the constants R and C (independently of the choice of x_0).

2.3 Construction of semi-positive metric on $f_*\omega_{X/Y}^{\otimes m}$

In order to finish the construction of semi-positive metric on $f_*\omega_{X/Y}^{\otimes m}$ we need the following lemma to eliminate the multiplier ideal sheaf $\mathcal{I}(h)$.

Proposition 2.5. Let $f : X \to Y$ be a surjective projective morphism with connected fibers between two complex manifolds. Suppose that $f_*\omega_{X/Y}^{\otimes m} \neq 0$ for some $m \geq 2$. (b) If h denotes the singular hermitian metric on $L = \omega_{X/Y}^{\otimes(m-1)}$ induced by the Narasimhan-Simha metric we just constructed, then

$$f_* \left(\omega_{X/Y} \otimes L \otimes \mathcal{I}(h)\right) \hookrightarrow f_*\omega_{X/Y}^{\otimes m}$$

is an isomorphism over the smooth locus of f.

PROOF IDEA 2.6. We prove that the natural inclusion is an isomorphism on $Y \setminus Z$ using the Ohsawa-Takegoshi theorem.

Hence we can transport the semi-positivity from $f_*(\omega_{X/Y} \otimes L \otimes \mathcal{I}(h))$ to the direct image of relative pluricanonical sheaves using the following lemma.

Lemma 2.7. Let $\phi : \mathscr{F} \to \mathscr{G}$ be a morphism between two torsion-free coherent sheaves that is generically an isomorphism. If $\mathscr{F}$ has a singular hermitian metric with semi-positive curvature, then so does $\mathscr{G}$.

Theorem 2.8 ([HPS18, Theorem 4.6])**.** Let $f : X \to Y$ be an algebraic fiber space. Suppose Y is compact.

(a) For any $m \in \mathbb{N}$, the torsion-free sheaf $f_*\omega_{X/Y}^{\otimes m}$ has a canonical singular hermitian metric with semi-positive curvature;
(b) If $c_1\left(\det f_*\omega_{X/Y}^{\otimes m}\right) = 0$ in $H^2(Y, \mathbb{R})$, $f_*\omega_{X/Y}^{\otimes m}$ is locally free, and the singular hermitian metric on it is smooth and flat;
(c) Every nonzero morphism $f_*\omega_{X/Y}^{\otimes m} \to \mathcal{O}_Y$ is split surjective.

To finish the proof of (b) we need the following two propositions.

Proposition 2.9 (Theorem of Katz and André)**.** Any coherent sheaf of $\mathcal{O}_X$ modules with integrable connection (flat connection) is locally free.

PROOF IDEA 2.10. The proof of the result requires the theory of D-modules. To be more precise.

3 A brief review of Kawamata's work on Iitaka conjecture when the base is of general type

In this section, we give a brief overview of Kawamata's result on Iitaka conjecture when the base is of general type. To keep the note to a reasonable length, we omit the proof of the results; all the details can be found in my note Iitaka Conjecture When base is of General Type.

3.1 Several Criteria for Albanese map to be a fibration

Theorem 3.1 (Albanese map as fibration when Kodaira dimension 0,[Kaw81, Theorem 1])**.** Let X be a non-singular and projective algebraic variety and assume that $\kappa(X) = 0$. Then the Albanese map $\alpha : X \to A(X)$ is an algebraic fiber space. In addition, if X has maximal Albanese dimension then the Albanese map is a birational contraction morphism.

3.2 Structure Theorem for Finite Covers of Abelian Varieties

Theorem 3.2 ([Kaw81, Theorem 13])**.** Let $f : X \to A$ be a finite morphism from a complete normal algebraic variety to an abelian variety.

Then $\kappa(X) \geq 0$ and there are an abelian subvariety B of A, étale covers $\tilde{X}$ and $\tilde{B}$ of X and B, respectively, and a complete normal algebraic variety $\tilde{Y}$ such that
(1) $\tilde{Y}$ is finite over A/B;
(2) $\tilde{X}$ is isomorphic to $\tilde{B} \times \tilde{Y}$;
(3) $\kappa(\tilde{Y}) = \dim \tilde{Y} = \kappa(X)$.

3.3 Kawamata's proof of Iitaka conjecture when base is of general type

Theorem 3.3 ([Kaw81, Theorem ??])**.** If $f : X \to Y$ is an algebraic fibre space with general fibre F and $\kappa(Y) = \dim Y$, then $\kappa(X) = \kappa(Y) + \kappa(F)$.

4 Proof of the Iitaka conjecture when the base has maximal Albanese dimension

Let us finish this note with the proof of the Iitaka conjecture when the base has maximal Albanese dimension.

Theorem 4.1. Let $f : X \to Y$ be an algebraic fiber space with general fiber F. Assume that Y has maximal Albanese dimension, then $\kappa(X) \geq \kappa(F) + \kappa(Y)$.

4.1 Reduce the problem to the case when the base is an abelian variety and $\kappa(X) = 0$

Proposition 4.2. To prove Theorem 4.1, it is sufficient to prove the case when $\kappa(X) = 0$ and Y is an abelian variety.

PROOF IDEA 4.3. We will divide the proof into 3 cases.
Case 1. When $\kappa(X) = -\infty$. The Green–Lazarsfeld–Simpson generic vanishing theorem is used. We assume by contradiction that $P_m(F) > 0$; this will imply that $f_*\omega_X^{\otimes m} \neq 0$. Consequently, we can apply the generic vanishing theorem developed in Section 1 to find some torsion point $P \in \mathrm{Pic}^0(A)$ with $h^0\left(X, \omega_X^{\otimes m} \otimes g^*P\right) = h^0\left(A, g_*\omega_X^{\otimes m} \otimes P\right) \neq 0$ and then

$$h^0\left(X, \omega_X^{\otimes km}\right) = h^0\left(X, \left(\omega_X^{\otimes m} \otimes P\right)^{\otimes k}\right) \neq 0,$$

which contradicts our assumption that $\kappa(X) = -\infty$.
Case 2. We assume that $\kappa(Y) = 0$. Using Theorem 3.1, we can reduce the base to an abelian variety Y. We then take an Iitaka fibration on the total space

$$h : X \to Z,$$

with general fiber G. We then restrict the fibration $f : X \to Y$ to a new fibration

$$G \to B',$$

with $\kappa(G) = 0$ and B' an abelian variety, and let H be the general fiber. This is exactly the situation of Proposition 4.1, giving

$$\kappa(G) \geq \kappa(H) + \kappa(B'),$$

thus $\kappa(H) = 0$.

$$\begin{tikzcd}
H = F \cap G \arrow[r, hook] \arrow[d, hook] & F \arrow[r] \arrow[d, hook] & h(F) \arrow[d, hook] \\
G \arrow[r, hook] \arrow[d] & X \arrow[r, "h"] \arrow[d, "f"] & Z \\
B \arrow[r, hook] & Y &
\end{tikzcd}$$

We then apply the easy addition formula to $F \to h(F)$, so that

$$\kappa(F) \leq \kappa(H) + \dim h(F) = \dim h(F),$$

which completes the proof of this case.
Case 3. When the base Y has maximal Albanese dimension. Then by Kawamata's structure theorem for finite covers of abelian varieties, we can decompose (an étale cover of) the base Y into a product of an abelian variety K and a variety of general type Z.

$$\begin{tikzcd}
F = E \cap H \arrow[r, hook] \arrow[d, hook] \arrow[dr] & E \arrow[r] \arrow[d, hook] & K \arrow[d] \\
H \arrow[r, hook] \arrow[d] & X \arrow[r, "p"] \arrow[d, "q"'] \arrow[dr, "f"] & K \\
Z \arrow[r] & Z & Y = Z \times K \arrow[l] \arrow[u]
\end{tikzcd}$$

The key point is to observe that the general fiber F of $f : X \to Y$ is also the general fiber of the induced map $E \to K$. We then apply the Iitaka conjecture when the base Z is of general type to get

$$\kappa(X) = \kappa(E) + \kappa(Z);$$

on the other hand, since K is an abelian variety, it is possible to prove the Iitaka conjecture for $E \to K$, giving

$$\kappa(E) \geq \kappa(F).$$

This completes the proof of the 3rd case. We can then apply the result of the Iitaka conjecture when the base is of general type to finish the proof.

4.2 Proof of the Iitaka Conjecture when the Base is an Abelian Variety and $\kappa(X) = 0$

We will finish the proof by showing that

Proposition 4.4. If $f : X \to A$ is an algebraic fiber space over an abelian variety, with $\kappa(X) = 0$, then we have

$$\mathscr{F}_m = f_* \omega_X^{\otimes m} \simeq \mathscr{O}_A$$

for every $m \in \mathbb{N}$ such that $H^0\left(X, \omega_X^{\otimes m}\right) \neq 0$.

PROOF IDEA 4.5. By Theorem 1.14 and Theorem 2.8, the direct image $\mathscr{F}_m$ is a flat unipotent vector bundle. Since $c_1(\det \mathscr{F}_m) = 0$ by Theorem 2.8(b), flatness and unipotency together force $\mathscr{F}_m \cong \bigoplus_{i=1}^r \mathcal{O}_A$. Using the assumption that $\kappa(X) = 0$, we have $r = 1$ and hence finish our proof.

Proof of Iitaka conjecture when base is an abelian variety and $\kappa(X) = 0$. By assumption, it is sufficient to prove that $\kappa(F) = 0$. Then, the result easily follows by Proposition 4.4. □

References

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